

TTX-MC138: Preclinically Promising yet Pharmacologically & Commercially Constrained

Benjamin Duong

April 2026

Disclosures	Page 3
Executive Summary	Page 3
Introduction	Page 4
microRNA-10b Biology	Page 6
Colorectal Cancer and miR-10b	
TTX-MC138	Page 10
Delivery System (IONP)	
Human Clinical Trials	Page 11
Phase 0 and Phase 1a	
Phase 2a	
Company Overview	Page 15
Intellectual Property	
Balance Sheet and Runway	
Summary and Conclusions	Page 16
References	Page 17

Disclosures: This paper was created for educational purposes based entirely on publicly available data and scientific literature, from the perspective of a non-licensed or accredited medical, financial, or investment professional. My views do not reflect any investment or scientific advice, and are personal views on the asset and company. I do not and will not hold any financial positions in the security (\$RNAZ), and do not have any personal relationships or financial ties to the company.

1. Executive Summary

TTX-MC138 is a miR-10b inhibitor delivered via iron oxide nanoparticles, currently in Phase 2a for ctDNA-positive colorectal cancer in an MRD setting. Phase 1a data shows no dose-limiting toxicities and early PK/PD activity, but raises several concerns. The measured plasma half-life of 19-25 hours combined with monthly dosing raises concerns about sustained target engagement, as the only publicly available pharmacodynamic measurement is 66% miR-10b inhibition at 24 hours post-infusion, with no PD data disclosed across the remainder of the 28-day dosing interval. This is concerning given that preclinical efficacy was demonstrated on a weekly dosing schedule, and the likely bottleneck to optimizing dose or frequency is the inherent liver hepatotoxicity and spleen-associated toxicity profile of the iron oxide nanoparticle delivery system. While Phase 1a demonstrated preliminary pharmacodynamic activity, all 3 CRC-enrolled patients experienced progressive disease under RECIST v1.1 or stable disease, providing a weak preliminary basis for CRC-specific clinical activity. The Phase 2a enrolls CRC post-curative-intent ctDNA-positive patients who lack the EMT-dominant biology that makes miR-10b inhibition mechanistically relevant. The proposed ctDNA clearance as a surrogate endpoint is mechanistically disconnected from TTX-MC138's proposed anti-metastatic mechanism, creating a trial design that cannot distinguish drug effect from background disease dynamics even in the event of a nominally positive signal. The core patent expires July 2032, and with cash runway ending Q4 2026 at the time of writing, the development timeline cannot reach approval within the patent exclusivity window. This paper argues that while the underlying miR-10b biology is scientifically credible, the current development program is misaligned with that biology that is structural and not correctable by protocol amendment.

2. Introduction

microRNA-10b (miR-10b) is an oncogenic microRNA that regulates tumor cell migration, invasion, and metastasis [1], with overexpression notably found in breast cancer, colorectal cancer, and glioblastoma [2][3]. Foundational preclinical efficacy data was generated in murine triple-negative breast cancer models (MDA-MB-231-luc-D3H2LN) using MN-anti-miR10b, the preclinical biologic from which TTX-MC138 is derived, developed in Dr. Medarova's lab at Massachusetts General Hospital. In vivo data showed promising results using a weekly dosing cycle protocol, with notable reduction of relative lymph node metastatic burden compared to molecular placebo (MN-SCR-miR) [4]. This dosing frequency will become relevant when compared to TTX-MC138's clinical half life.

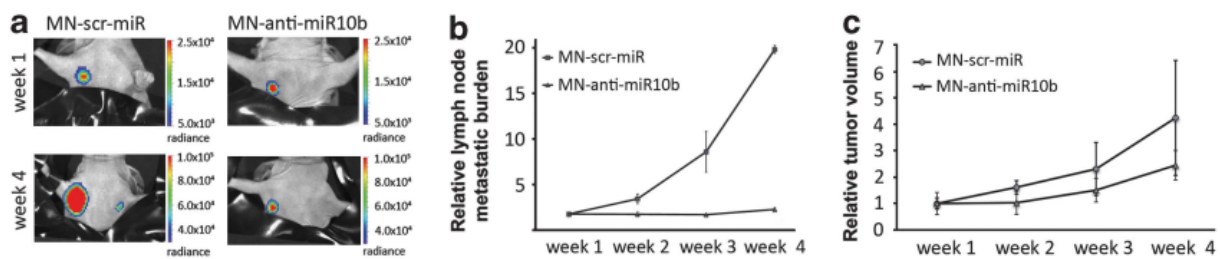


Figure 1. MN-anti-miR10b vs inactive control in murine models [4]

Following the same journal, Dr. Medarova and her team noted that there **wasn't any observable apoptosis in the primary tumor and lymph node**, but tumor cell proliferation only in the lymph node decreased ($p < 0.0026$) [4]. Without apoptotic activity within the primary tumor and confinement of antiproliferative effects to lymph nodes, these results have implications toward TTX-MC138's clinical endpoint interpretations discussed further in the clinical trials section.

Figure 7.

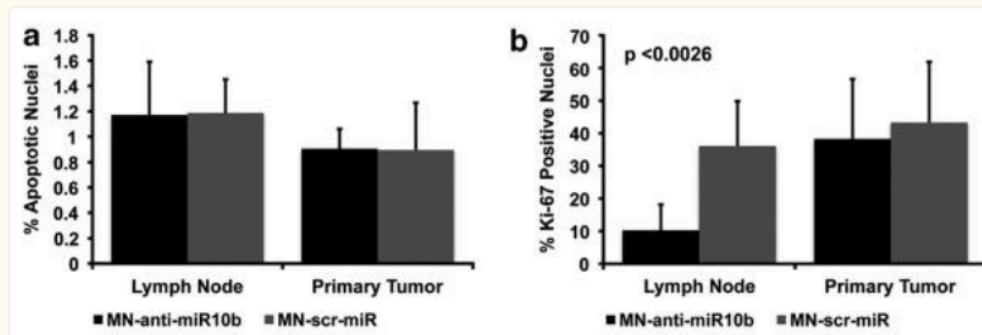


Figure 2. MN-anti-miR10b vs inactive control apoptotic and antiproliferative effects in vivo [4]

With this preclinical data, Transcode Therapeutics was founded on collaborations between Dr. Anne Moore and Dr. Zdravka Medarova (Chief Scientific Officer) where they developed the first iterations of the TTX platform published in 2006 [5] and their commercialized iteration along with “MN-miR-10b” (TTX-MC138) patented in 2012 [4][6]. TTX-MC138 subsequently entered a Phase 0 microdose trial, and in the last few months, completed a Phase 1a clinical trial to test for safety (TEAEs) and early PK data as their primary endpoints with anti-tumor activity as a secondary endpoint (NCT06260774). As of April 2026, TTX-MC138 is in a Phase 2a clinical trial in an MRD setting for CRC patients partnered with Quantum Leap under its adaptive platform trial infrastructure named “Pre I-SPY Platform Program” (NCT05868226). The mechanistic rationale for CRC application rests on miR-10b's independently documented role in colorectal invasion and metastasis, distinct from the breast cancer models used for initial efficacy testing.

The question this paper addresses is not whether miR-10b represents a valid therapeutic target, the preclinical literature suggests it does, under the right conditions. The question is whether TTX-MC138, as currently administered, in the population currently enrolled, backed by a company with significant financial and IP constraints, can generate a clinically valuable and commercially viable result. The below sections examine the pharmacological, clinical, and commercial evidence bearing on that question.

3. microRNA-10b

microRNA-10b is upregulated by TWIST-1 and HIF-1 α , but, when overexpressed, downregulates tumor suppressors including BIM, KLF4, MICB, and HOXD10, while indirectly upregulating the prometastatic gene RhoC [3][7][8][9]. The below sections will explore miR-10b's connection with colorectal cancer metastasis.

3.1 Upstream Regulators: TWIST-1, HIF-1 α , CpG DNA Methylation

miR-10b's most well documented upstream regulator is TWIST-1, a transcription factor that forms homo and hetero dimers around Nde1 E-box to activate or repress target genes, regulating miR-10b's transcription [10]. HIF-1 α acts as an indirect upstream regulator through a sequential axis: hypoxic conditions stabilize HIF-1 α , inducing TWIST-1 expression, which in turn drives miR-10b upregulation [10][11][12].

In context of TTX-MC138's monthly dosing schedule, HIF-1 α -mediated TWIST-1 induction in hypoxic tumor regions represents a continuous upstream re-activation signal that operates independently of miR-10b inhibition. TWIST-1 has been associated with accelerating the EMT process and promoting lymph node and distant metastasis, and is capable of inducing EMT independently [3][10] [13]. However, miR-10b expression is transcriptionally dependent on TWIST-1, placing it downstream in this EMT-regulatory axis [13]. In one study, **CpG DNA Methylation of miR-10b's allele has been found to result in low miR-10b expression in 46% of colorectal cancer tumors** [3]. Transcode Therapeutics has **not specified any baseline miR-10b expression filters in their Phase 2a key inclusion criteria**, meaning that a significant subpopulation in their enrolled patients may have low or absent target expression.

3.2 Downstream Targets: HOXD10, MICB, KLF4, BIM

HOXD10, a tumor suppressor gene that prevents tumor cell migration and invasion, is targeted by miR-10b through translational inhibition which upregulates RhoC, a metastatic promoting gene. Upregulated miR-10b levels are associated with characteristics of aggressive, high migratory metastasis [14][15]. However, as the HOXD10-RhoC axis primarily drives the initial departure of tumor cells from the primary site following EMT [18], its relevance in the post-curative-intent ctDNA-positive population, where dissemination has already occurred, may be limited to earlier stages of the metastatic process. TransCode has not disclosed TNM staging criteria, CMS subtype, or baseline miR-10b expression levels for enrolled patients, leaving the question of whether TTX-MC138 is being administered within miR-10b's active biological window unresolved.

MICB, a ligand used for NK-cell activation, has been shown to be downregulated by miR-10b, preventing NK cell recognition via the NKG2D receptor [8]. MiR-10b's suppression of MICB occurs within a broader network of MICB-targeting miRNAs (miR-520d-5p, miR-106b, miR-17, miR-20a, and miR-93) that also regulate TWIST1 expression, suggesting that miR-10b's metastatic and cytotoxicity escape functions are connected to a network of miRNAs [8]. This redundancy suggests that inhibiting miR-10b alone via TTX-MC138 may be insufficient to restore MICB surface expression and NK cell-mediated cytotoxicity, as the remaining network members continue to suppress MICB independently.

KLF4, a tumor suppressor gene in colorectal cancer preventing cell proliferation, migration, and invasion was found to be downregulated by miR-10b [16][17]. Suppressing KLF4 has been found to decrease E-cadherin (a cell-cell adhesion mediator) and increase Snail expression, regulators of EMT, thus linking miR-10b overexpression to aggressive, metastasized colorectal cancer [16].

BIM, a pro-apoptotic mediator of chemotherapy induced cell death, has been found to play a critical role in chemosensitivity in 5-FU [9]. However, BIM mRNA in CRC was found to be inhibited through miR-10b post-transcriptionally, thus showing BIM as a downstream target of miR-10b in the context of CRC [9]. With BIM suppression, CRC may be more resistant to chemotherapy, making miR-10b an indicator towards refractory CRC patients.

4. Colorectal Cancer Metastasis

In colorectal cancer metastasis, the EMT process is induced through the loss of tight junctions, adherens junctions, and desmosomes (cell-cell adhesion) where CRC cells are able to invade into nearby regional tissue and acquire a mesenchymal phenotype as circulating tumor cells (CTCs) which subsequently extravasate and colonize distant sites through MET-driven establishment of micrometastases [19]. It is important to note that the EMT/MET axis is transient, leading to partial EMT and reversibility among CTCs, thus making CRC difficult to treat due to its heterogeneity and phenotypic plasticity across the EMT/MET spectrum [19] [22].

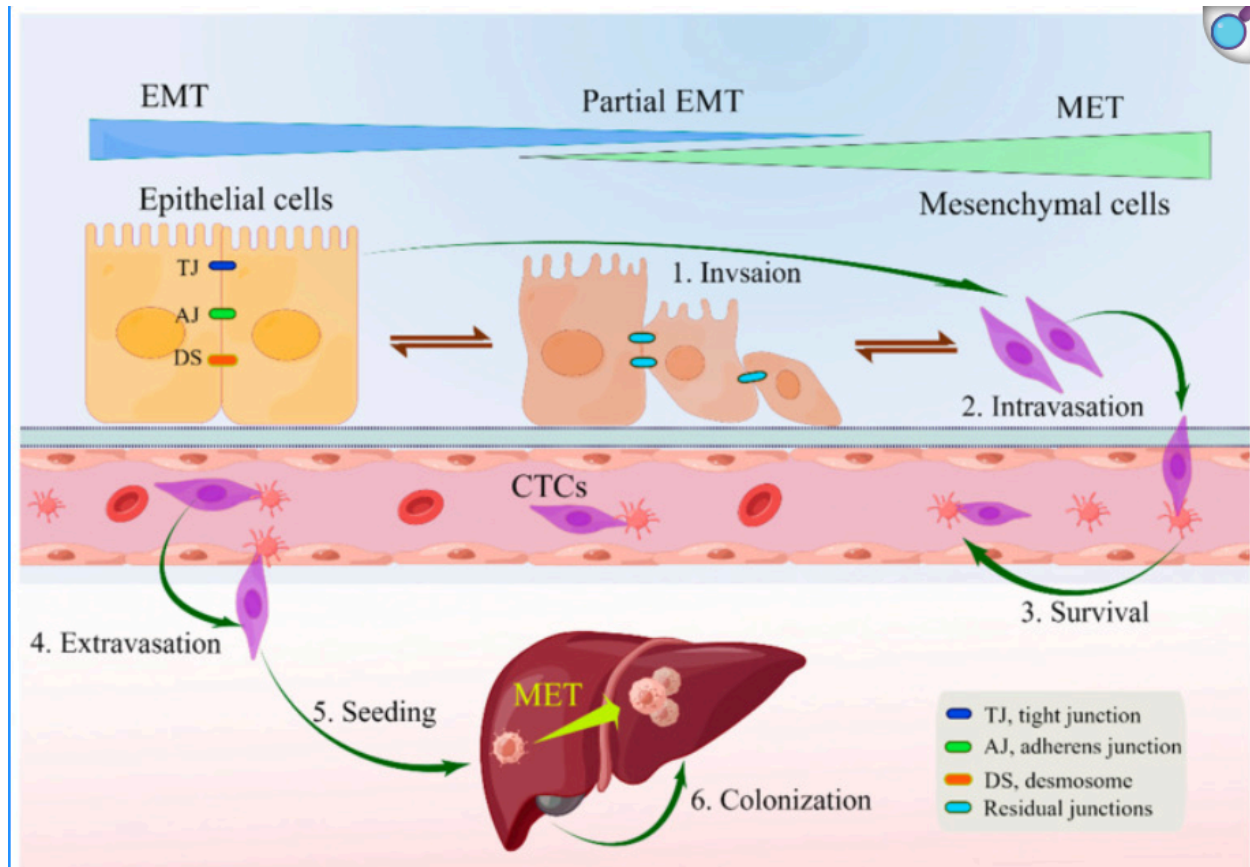


Figure 3. EMT/MET colorectal cancer metastasis process [19]

4.1 Colorectal Cancer Metastasis and miR-10b

As previously mentioned, ~46% of CRC samples have low miR-10b expression through CpG DNA methylation, and its expression has been found to only be **slightly elevated in primary CRC tumors compared to normal intestinal tissue** [18]. Low miR-10b expression has been reported in liver metastasis from colorectal cancer, but is primarily expressed in TNM II and III suggesting that mir-10b is most biologically active right at early lymph node involvement but does not sustain metastatic establishment [18] [20]. A study characterizing

Stage II CRC as a transitional phase during which micrometastatic volume in lymph nodes continuously increases suggests that active dissemination, and by extension miR-10b-driven EMT, is most significant at this stage, prior to the establishment of detectable lymph node metastasis in Stage III [21].

active	Tx	T0	Tis	T1	T2	T3	T4	Nx	N0	N1	N2	N3	M0	M1
maybe									no node+		too late	too late	too late	liver metastasis
too late		T1-T4 represents tumor size increase (not indicative of LNM)												
										miR-10b active, but some have already CTCs already colonized				

Figure 4. miR-10b activity based on T, N, M stages

5. TTX-MC138:

TTX-MC138 is an antisense oligonucleotide biologic using iron oxide nanoparticles as the delivery method to inhibit miR-10b expression. MGH's foundational patent exclusively licensed to TransCode Therapeutics (US9,629,812) covers its preclinical formulation and data [6]. Interestingly, all three patents' backgrounds begin with detailing breast cancer rather than colorectal cancer. Within their TTX-MC138 IP section, they do not explicitly state the exact composition of their nanoparticles (ie magnetite, manganese ferrite, maghemite etc.), however, they do disclose that they are manufactured using co-precipitation, a commonly used method to synthesize iron oxide particles [23].

5.1 Iron Oxide Nanoparticle Delivery System (TTX Platform) Tox Concerns

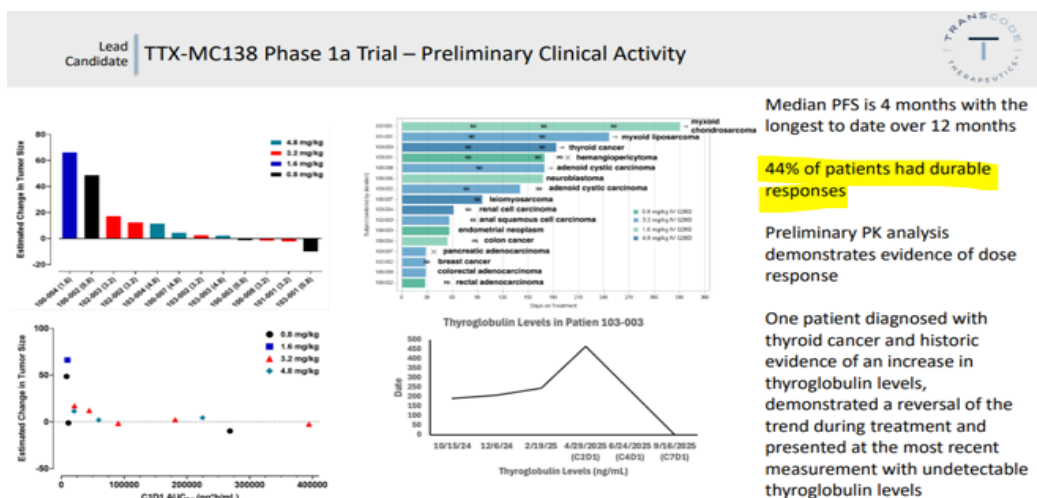
While Transcode has not specified the exact size of the clinical TTX-MC138 formulation, there are noted tox concerns within literature on iron oxide nanoparticles (IONP), potentially implicating TTX-MC138. First, IONPs around 10nm that were spherical were found to accumulate in the liver and spleen, later penetrating the brain and uterus, while IONPs with higher dimensions were found to accumulate in the kidneys [23]. Additionally, IONPs reduced surface cell marker (CD40 and MHC class II molecule) expressions and number of lymphocytes in the blood [23]. However, we can infer that the clinical formula of TTX-MC138 falls within the smaller size ranges specified in the patent, as their Phase 0 results mentioned that the highest concentrations of TTX-MC138 at 24 hours were found primarily in the liver and spleen [25]. Surveillance of Transcode's Phase 2a trial of liver hepatotoxicities as well as ALT, AST, and ALP common hepatotoxicity markers, will be important to watch [26]. Transcode did not mention if they measure lymphocyte count via CBC or any reduced surface cell marker expressions in their clinical trials. If lymphocyte counts are in fact decreased in the blood with decreased surface cell marker expression for antigen presentation and T cell activation, this implicates immune responses for controlling residual disease (CRC) in an MRD setting.

6. Human Clinical Trials:

Transcode began their human clinical trial process in 2023 with their first patient with stage IV metastatic breast cancer without any toxicity concern, and has since started their Phase 2a trial in 1H 2026.

6.1 Phase 0 and Phase 1a Results

Transcode announced their Phase 0 results with a half life of 18.7 hours at a 100 μ g microdose over the course of 5 months (NCT05908773) [27]. However, the Phase 1a results came with some questionable IR/PR moments, and did not release any tissue PK data. First, on their investor's presentation slide, they announced that 44% of patients had *durable responses*, while announcing to the media/showing on their preliminary PD graph that 44% of patients had stable disease which falls under RECIST v.1 likely to manage investor expectations. Out of the 12 evaluated patients, 10 had stable disease, and 2 had progressive disease. Notably, 2 CRC patients had progressive disease and 1 had stable disease. Not only that, they stated that the median PFS was 4 months long (time until progression or death during or after treatment) while stating that median treatment duration was also 4 months. Under standard RECIST 1.1 criteria, stable disease does not qualify as an objective response, and the ORR (PR + CR only) is not explicitly reported [28]. The conflation of disease control with clinical response overstates the efficacy signal and should be clarified before drawing conclusions about therapeutic activity.



BOSTON, Oct. 14, 2025 /PRNewswire/ -- TransCode Therapeutics, Inc. (NASDAQ: RNAZ) announces completion of Phase 1a clinical trial with TTX-MC138, an investigational inhibitor of microRNA-10b, or miR-10b. By meeting its primary safety endpoint and defining a recommended Phase 2 dose, TTX-MC138 is moving forward into the next stage of clinical evaluation to assess its efficacy across selected metastatic diseases and for multiple indications.

Primary objectives of the trial focused on safety, tolerability, pharmacokinetics ("PK") and establishment of a Phase 2 dose (RP2D). A total of sixteen patients were treated across four escalating dose levels. No significant treatment-related safety events or dose limiting toxicities were observed.

TTX-MC138 was administered to 16 patients (77 total doses to date) with positive pharmacodynamic effects over all four administered dose range levels. Currently three patients remain on trial receiving TTX-MC138. The median treatment duration was four months. Importantly, the duration of treatment for all patients ranged from two to 12 cycles, indicative of tolerability and disease control. Forty-four per cent or seven out of sixteen patients were classified as having stable disease lasting 4 months or longer. Preliminary data in 16 patients showed positive pharmacodynamic effects over a wide dose range, consistent with preclinical models and TransCode's Phase 0 clinical trial. Of note, one patient diagnosed with thyroid cancer and historic evidence of an increase in thyroglobulin levels, demonstrated a reversal of the trend during treatment and presented at the most recent measurement with undetectable thyroglobulin levels.

Study investigator William McKean, MD, PhD (The START Center for Cancer Research) stated, "The clinical benefit we are observing in patients treated with TTX-MC138 is compelling given the pre-clinical data and length of time the drug remains in the tumor cells".

Data analysis and monitoring are ongoing. A final clinical study report, scientific presentations and publications are planned.

Analyzing the Phase 1a trial's PK behaved like a two-compartment system, where there was a rapid distribution phase from blood plasma into tissue followed by a slower terminal elimination plasma phase driven by redistribution into the tissue [29] [30] [31]. In all 4 cohorts (0.8mg/kg, 1.6mg/kg, 3.2mg/kg, 4.8mg/kg), Transcode provided the blood plasma PK in its investor presentation slides, in which each respective cohort's C_{pt} were: $8000 \cdot e^{(-0.2327t)} + 38.2 \cdot e^{(-0.01011t)}$, $30000 \cdot e^{(-0.2525t)} + 165.5 \cdot e^{(-0.03587t)}$, $75000 \cdot e^{(-0.2802t)} + 182.4 \cdot e^{(-0.02943t)}$, and $90000 \cdot e^{(-0.2257t)} + 785.1 \cdot e^{(-0.02809t)}$.

PK analysis on publicly available plasma concentration data indicates the C_{p0} (the theoretical plasma concentration immediately after IV infusion) scales measured across cohorts 1.6 - 4.8 mg/kg showed signs of dose proportional exposure, and is backed by consistent dose dependent V_c values (the theoretical volume of distribution into the blood and tissue) between 0.0425 - 0.0530 (L/kg) suggesting that TTX-MC138 is mainly confined in the vascular component [28]. The 0.8 mg/kg cohort showed inconsistent dose dependency and broader distribution with V_c of 0.099 (L/kg), V_{ss} of 0.286 (L/kg), MRT of 13.67 hours, CL of 0.0209 (L/h/kg), and $T_{1/2\beta}$ of 68.58 hours, likely reflecting unsaturated uptake of TTX-MC138 at low therapeutic plasma exposure and higher clearance relative to dose.

Furthermore, the V_{ss} values within the 1.6 - 4.8mg/kg cohorts are 0.049 - 0.072 (L/kg) and stable CL range of 0.011 - 0.013 (L/h/kg), confirming that TTX-MC138 mainly stays within the blood plasma and has minimal peripheral distribution with consistent elimination. Together, these parameters show an MRT of 4.23 - 6.47 hours, indicating that drug molecules spend less than 7 hours in systemic circulation on average before uptake or elimination, consistent with time-limited systemic exposure. However, no information on tissue PK was made publicly available, which would be required to confirm whether clearance reflects target distribution.

Terminal half-life ($T_{1/2\beta}$) across the 1.6–4.8 mg/kg cohorts ranged from 19.3–24.7 hours. Measurable plasma concentrations confirmed at 168 hours post-infusion suggest that the half-life of miR-10b suppression likely exceeds plasma terminal half-life. Pharmacodynamic analysis confirmed 66% inhibition of miR-10b at 24 hours post-infusion in patients with high baseline miR-10b expression, directly replicating the Phase 0 observation and establishing proof-of-mechanism target engagement. However, no further PD data past 24 hours was shared, leaving the gap of sustained target engagement across the full 28 day dosing cycle unresolved. In their preclinical mouse models, sustained suppression over the full dosing period was a prerequisite for the mechanism to be relevant [4]. The preclinical efficacy data was generated on a weekly dosing schedule, and TransCode has not published evidence that monthly dosing achieves equivalent or sufficient target coverage to replicate those conditions. The bottleneck in either increasing dosages or increasing dosing frequency is in the inherent tox risks associated with iron oxide nanoparticles; liver hepatotoxicity and spleen-associated tox risks.

Further, 10/12 patients achieved only stable disease by RECIST criteria, suggesting that while miR-10b inhibition may demonstrate cytostatic activity, it does not produce meaningful cytotoxic or tumor-shrinking effects in refractory metastatic or failed prior lines of therapy. This is mechanistically consistent with miR-10b's established role in regulating invasion and metastatic dissemination rather than tumor proliferation directly; inhibiting miR-10b would be expected to produce cytostatic rather than cytotoxic effects. Clinically, for failed prior lines of therapy patients, does cytostatic activity warrant the use of TTX-MC138, where other current and emerging treatments (ADCs, mAbs, and bispecific antibodies) have been demonstrated meaningful tumor regression? Based on the early PK/PD evidence, TTX-MC138 does not prove itself as a viable late-line therapeutic option, a conclusion Transcode implicitly acknowledges by shifting their Phase 2a patient pool towards an MRD post-curative-intent ctDNA-positive pool.

Cohorts:	0.8	1.6	3.2	4.8
a; c (0h)	8000	30000	75000	90,000
c (24h)	30	70	90	400
alpha; Cp	0.23275	0.252519	0.28022641	0.22567085
b (y-int)	38.2	165.5	182.4	785.1
beta; Cp-C	0.010106	0.035867	0.0294267	0.028094128
T-alpha	2.978076	2.74493	2.47352558	3.071496298
T-beta	68.58659	19.32572	23.5550395	24.67231538
Cp0	8038.2	30165.5	75182.4	90785.1
MRT (avg time in blood)	13.67423632	4.854456412	4.256979691	6.471909514
Vc (L/kg)	0.099524769	0.053040725	0.042563153	0.052872112
Vss	0.286526113	0.062923029	0.049743801	0.072794558
Cl	0.020969021	0.0129642	0.0116857	0.011247638

6.2 Phase 2a: Will it work in MRD?

CRC metastasis proceeds through a reversible EMT/MET axis where primary tumor cells undergo EMT to enable invasion and intravasation, then revert through MET to colonize and establish micrometastases at distant sites [19]. This suggests that TTX-MC138's mechanism of action, preventing EMT by inhibiting miR-10b, may be most pharmacologically relevant in early stages of TNM, prior to resection, rather than in the post-curative-intent MRD setting where primary tumor dissemination has already occurred via EMT and micrometastatic seeding is the dominant biological process [19]. Furthermore, ctDNA positivity reflects shed primarily through tumor cell apoptosis or necrosis, rather than from actively invasive circulating tumor cells. This raises questions on the relevance of inhibiting active EMT. This further questions the relevance of the HOXD10-RhoC axis which operates in the context of active EMT and intravasation in a population where active invasion is no longer the dominant process [19].

In the post-curative-intent MRD setting, TTX-MC138 faces a dual biological problem: the colonized micrometastatic cells have already completed MET and naturally depleted miR-10b, leaving no active target to inhibit, and the primary tumor source of new EMT-driven dissemination has been surgically removed, eliminating the upstream pipeline TTX-MC138 is designed to interrupt. Publicly available data on TransCode's Phase 2a clinical trial inclusion criteria specify ctDNA positivity in post-curative-intent CRC patients who are radiographically NED, a patient population whose disease stage falls outside the biological window in which miR-10b expression has been shown to be active. This does not support confidence in the positive outcome of the trial, given the short biological window that miR-10b may have the highest chances for success in.

7. Company Financial and IP-Commercial Positions

Transcode IPO'd in 2021, only raising \$25,000,000 gross at \$4.00 a share and offering 6,250,000 shares [32]. Since its IPO, it has had a YoY cash burn for operating activities of between \$13,000,000 to over \$19,500,000 [33][34]. As of December 31, 2025, they reported to have around \$17,800,000 after placing a private placement of their Series B shares for \$20,000,000 with a \$5,000,000 promissory note, and noted in their 2025 10-K that they believed this would only last them until Q4 2026. This effectively heavily diluted their shares, giving around 91% of the company to DEFJ and leaving their prior shareholders with less equity in the company. Clinical trials are known to be one of the most expensive aspects within a medicine's lifetime, and with only \$17.8M, Transcode is financially hanging by a thread and its continuation lies heavily on their Phase 2a trial for TTX-MC138. From a pipeline-portfolio management standpoint, having a single asset be the determining factor for the company's clinical and financial success without other notable assets presents a strong risk for investors and the company.

Another major risk factor as previously mentioned, is their IP constraints regarding TTX-MC138. Currently they have not completed their Phase 2a trial yet, and with only 6 years left before patent expiry, Transcode almost certainly does not have the timeframe to successfully commercialize, under the assumption that it receives NDA and BLA approval after Phase 3 results. Additionally, they are ineligible for Orphan Disease Status, which would allow them 7 extra years of market exclusivity as mCRC is not a rare disease.

8. Summary and Conclusions

miR-10b as a therapeutic target holds many preclinical promises as a prognostic indicator for cancer metastasis, but ultimately presents many risks being connected into an unknown network of other microRNAs [35]. Prior attempts at making microRNAs have largely failed, largely for this reason as well, but Transcode is betting that miR-10b is a good therapeutic candidate (in the MRD setting with CRC patients). Based on the mCRC and miR-10b biology and mechanisms, TTX-MC138 PK/PD evidence, and Phase 2a trial design, all axes point towards low confidence of success that Transcode is betting on.

A better avenue for miR-10b, a validated prognostic factor in cancer metastases, may be in diagnostics. As previously discussed, the active biological window for miR-10b starts long before distant metastasis, therefore, early detection of miR-10b may provide physicians a better insight into each individual patients' treatment plan. While not specific to miR-10b, some microRNA biomarkers are being used in diagnostics, and UCLA Professor Dr. Joanne Weidhaas's company MiraDX uses germline biomarker tests based on microRNAs. However, reproducibility of microRNA expression across patient samples remains an open challenge [36]. It is clear that as a therapeutic target, miR-10b has fundamental risks that make it undesirable, but as a diagnostic biomarker, the case is more compelling.

References:

- [1] Tang, J., Ahmad, A., & Sarkar, F. H. (2012). The role of micrnas in breast cancer migration, invasion and metastasis. *International Journal of Molecular Sciences*, 13(10), 13414–13437. <https://doi.org/10.3390/ijms131013414>
- [2] Gabriely, G., Teplyuk, N. M., & Krichevsky, A. M. (2011). Context effect: Microrna-10b in cancer cell proliferation, spread and death. *Autophagy*, 7(11), 1384–1386. <https://doi.org/10.4161/auto.7.11.17371>
- [3] Abdelmaksoud-Dammak, R., Chamtour, N., Triki, M., Saadallah-Kallel, A., Ayadi, W., Charfi, S., Khabir, A., Ayadi, L., Sallemi-Boudawara, T., & Mokdad-Gargouri, R. (2017). Overexpression of mir-10b in colorectal cancer patients: Correlation with *twist-1* and E-cadherin expression. *Tumor Biology*, 39(3), 101042831769591. <https://doi.org/10.1177/1010428317695916>
- [4] Yigit, M. V., Ghosh, S. K., Kumar, M., Petkova, V., Kavishwar, A., Moore, A., & Medarova, Z. (2012). Context-dependent differences in mir-10b breast oncogenesis can be targeted for the prevention and arrest of lymph node metastasis. *Oncogene*, 32(12), 1530–1538. <https://doi.org/10.1038/onc.2012.173>
- [5] Medarova, Z., Pham, W., Kim, Y., Dai, G., & Moore, A. (2006). *in vivo* imaging of tumor response to therapy using a dual-modality imaging strategy. *International Journal of Cancer*, 118(11), 2796–2802. <https://doi.org/10.1002/ijc.21672>
- [6] Medarova, Z., Yigit, M. V., Moore, A., & Corp, G. H. (2011, July 22). *US9629812B2 - Therapeutic nanoparticles and methods of use thereof - Google Patents*. <https://patents.google.com/patent/US9629812B2/en>
- [7] Sheedy P, Medarova Z. The fundamental role of miR-10b in metastatic cancer. *Am J Cancer Res*. 2018 Sep 1;8(9):1674-1688. PMID: 30323962; PMCID: PMC6176190.
- [8] Tsukerman, P., Yamin, R., Seidel, E., Khawaled, S., Schmiedel, D., Bar-Mag, T., & Mandelboim, O. (2014). Mir-520D-5p directly targets Twist1 and downregulates the metastamir mir-10b. *Oncotarget*, 5(23), 12141–12150. <https://doi.org/10.18632/oncotarget.2559>
- [9] Nishida, N., Yamashita, S., Mimori, K., Sudo, T., Tanaka, F., Shibata, K., Yamamoto, H., Ishii, H., Doki, Y., & Mori, M. (2012). MicroRNA-10B is a prognostic indicator in colorectal cancer and confers resistance to the chemotherapeutic agent 5-fluorouracil in colorectal cancer cells. *Annals of Surgical Oncology*, 19(9), 3065–3071. <https://doi.org/10.1245/s10434-012-2246-1>

- [10] Qin, Q., Xu, Y., He, T. *et al.* Normal and disease-related biological functions of Twist1 and underlying molecular mechanisms. *Cell Res* 22, 90–106 (2012).
<https://doi.org/10.1038/cr.2011.144>
- [11] Yang, M.-H., Wu, M.-Z., Chiou, S.-H., Chen, P.-M., Chang, S.-Y., Liu, C.-J., Teng, S.-C., & Wu, K.-J. (2008). Direct regulation of twist by HIF-1 α promotes metastasis. *Nature Cell Biology*, 10(3), 295–305. <https://doi.org/10.1038/ncb1691>
- [12] Shen, G., Li, X., Jia, Y., Piazza, G. A., & Xi, Y. (2013). Hypoxia-regulated micrnas in human cancer. *Acta Pharmacologica Sinica*, 34(3), 336–341.
<https://doi.org/10.1038/aps.2012.195>
- [13] Ma, L., Teruya-Feldstein, J., & Weinberg, R. A. (2007). Tumour invasion and metastasis initiated by microRNA-10B in breast cancer. *Nature*, 449(7163), 682–688.
<https://doi.org/10.1038/nature06174>
- [14] Surendran, H., Palaniyandi, T., Natarajan, S., Hari, R., Viwanathan, S., Baskar, G., Abdul Wahab, M. R., Ravi, M., & Rajendran, B. K. (2023). Role of homeobox D10 gene targeted signaling pathways in cancers. *Pathology - Research and Practice*, 248, 154643.
<https://doi.org/10.1016/j.prp.2023.154643>
- [15] WANG, Y., LI, Z., ZHAO, X., ZUO, X., & PENG, Z. (2016). Mir-10B promotes invasion by targeting hoxd10 in colorectal cancer. *Oncology Letters*, 12(1), 488–494.
<https://doi.org/10.3892/ol.2016.4628>
- [16] Ma, Y., Wu, L., Liu, X., Xu, Y., Shi, W., Liang, Y., Yao, L., Zheng, J., & Zhang, J. (2017). KLF4 inhibits colorectal cancer cell proliferation dependent on NDRG2 signaling. *Oncology Reports*, 38(2), 975–984. <https://doi.org/10.3892/or.2017.5736>
- [17] Xie, Y., Zhao, J., Liang, Y., Chen, M., Luo, Y., Cui, X., Jiang, B., Peng, L., & Wang, X. (2019). MicroRNA-10B controls the metastasis and proliferation of colorectal cancer cells by regulating krüppel-like factor 4. *Artificial Cells, Nanomedicine, and Biotechnology*, 47(1), 1722–1729. <https://doi.org/10.1080/21691401.2019.1606006>
- [18] Strubberg, A. M., & Madison, B. B. (2017). MicroRNAs in the etiology of colorectal cancer: Pathways and clinical implications. *Disease Models & Mechanisms*, 10(3), 197–214. <https://doi.org/10.1242/dmm.027441>
- [19] Lu, J., Kornmann, M., & Traub, B. (2023). Role of epithelial to mesenchymal transition in colorectal cancer. *International Journal of Molecular Sciences*, 24(19), 14815.
<https://doi.org/10.3390/ijms241914815>

- [20] Hur, K., Toiyama, Y., Schetter, A. J., Okugawa, Y., Harris, C. C., Boland, C. R., & Goel, A. (2015). Identification of a metastasis-specific MicroRNA signature in human colorectal cancer. *JNCI: Journal of the National Cancer Institute*, 107(3). <https://doi.org/10.1093/jnci/dju492>
- [21] Yamamoto, H. (2022). Micrometastasis in lymph nodes of colorectal cancer. *Annals of Gastroenterological Surgery*, 6(4), 466–473. <https://doi.org/10.1002/ags3.12576>
- [22] Cañellas-Socias, A., Sancho, E., & Batlle, E. (2024). Mechanisms of metastatic colorectal cancer. *Nature Reviews Gastroenterology & Hepatology*, 21(9), 609–625. <https://doi.org/10.1038/s41575-024-00934-z>
- [23] Baabu, P. R., Kumar, H. K., Gumpu, M. B., Babu K, J., Kulandaisamy, A. J., & Rayappan, J. B. (2022). Iron oxide nanoparticles: A review on the province of its compounds, properties and biological applications. *Materials*, 16(1), 59. <https://doi.org/10.3390/ma16010059>
- ~~[24] Medarova, Z., Yigit, M. V., & Moore, A. (2017). *Therapeutic nanoparticles and methods of use thereof* (U.S. Patent No. 9,629,812). U.S. Patent and Trademark Office~~
- [25] Medarova, Z., Robertson, N., Ghosh, S., Duggan, S., & Varkaris, A. (2024). Initial clinical experience with the first-in-class anti-metastasis therapeutic TTX-MC138. *Journal of Clinical Oncology*, 42(16_suppl). https://doi.org/10.1200/jco.2024.42.16_suppl.e15072
- [26] Thakur, S., Kumar, V., Das, R., Sharma, V., & Mehta, D. K. (2024). Biomarkers of hepatic toxicity: An overview. *Current Therapeutic Research*, 100, 100737. <https://doi.org/10.1016/j.curtheres.2024.100737>
- [27] Medarova, Z., Fu, S., Varkaris, A., McKean, M., Barve, M., Spira, A., Vlock, D., Fortin, L., & Dugan, S. (2026). Abstract PS1-09-17: Phase 1 clinical testing of a first-in-class antisense oligonucleotide therapeutic against advanced solid tumors of multiple tissue origins. *Clinical Cancer Research*, 32(4_Supplement). <https://doi.org/10.1158/1557-3265.sabcs25-ps1-09-17>
- [28] Eisenhauer, E. A., Therasse, P., Bogaerts, J., Schwartz, L. H., Sargent, D., Ford, R., Dancey, J., Arbuck, S., Gwyther, S., Mooney, M., Rubinstein, L., Shankar, L., Dodd, L., Kaplan, R., Lacombe, D., & Verweij, J. (2009). New response evaluation criteria in solid tumours: Revised recist guideline (version 1.1). *European Journal of Cancer*, 45(2), 228–247. <https://doi.org/10.1016/j.ejca.2008.10.026>
- [29] Gabrielsson, J., & Weiner, D. (2016). *Pharmacokinetic and pharmacodynamic data analysis: Concepts and applications* (5th ed.). Swedish Pharmaceutical Press. [1, 2]
- [30] Stark, J. (n.d.). *Two compartment body model and Vd terms*. College of Pharmacy, University of Florida. <https://pharmacy.ufl.edu/files/2013/01/two-compartment-model.pdf>

- [31] Wagner, J. G. (1983). Significance of ratios of different volumes of distribution in pharmacokinetics. *Biopharmaceutics & Drug Disposition*, 4(3), 263–270.
<https://doi.org/10.1002/bdd.2510040307>
- [32]<https://ir.transcodetherapeutics.com/2021-07-08-TransCode-Therapeutics,-Inc-Announces-Pr-icing-of-Initial-Public-Offering>
- [33]
<https://app.quotemedia.com/data/downloadFiling?webmasterId=101533&ref=319955461&type=HTML&symbol=RNAZ&cdn=5d1b8c7430c083508e1c22e158a2f56c&companyName=TransCo de+Therapeutics+Inc.&formType=10-K&dateFiled=2026-04-15>
- [34]
<https://www.sec.gov/Archives/edgar/data/1829635/000155837025004843/rnaz-20241231x10k.htm>
- [35] Zhang, S., Cheng, Z., Wang, Y., & Han, T. (2021). The risks of Mirna Therapeutics: In A drug target perspective. *Drug Design, Development and Therapy*, Volume 15, 721–733.
<https://doi.org/10.2147/dddt.s288859>
- [36] El-Daly, S. M., Gouhar, S. A., & Abd Elmageed, Z. Y. (2023b). Circulating micrnas as reliable tumor biomarkers: Opportunities and challenges facing clinical application. *The Journal of Pharmacology and Experimental Therapeutics*, 384(1), 35–51.
<https://doi.org/10.1124/jpet.121.000896>

